Lines and Line Segments in 3-Space

In 2-space a line L is given by two pieces of data: a point and a slope. In 3-space, we also need two pieces of data: a point $P_0(x_0, y_0, z_0)$ and a *vector* $\mathbf{v}$ pointing in the direction of the line:

Vector Equation for a Line. A vector equation for the line L through $P_0(x_0, y_0, z_0)$ parallel to $\mathbf{v}$ is

$$\mathbf{r}(t) = \mathbf{r}_0 + t\mathbf{v}, \qquad \text{where} \qquad -\infty < t < \infty.$$

Here $\mathbf{r}$ is the position vector of a point $P(x, y, z)$ on L and $\mathbf{r}_0$ is the position vector of $P_0(x_0, y_0, z_0)$.

Since the above is a vector equality, their respective components are equal. Since

$$\mathbf{r}(t) = \langle x(t), y(t), z(t)\rangle, \qquad \mathbf{r}_0 = \langle x_0, y_0, z_0\rangle, \qquad \text{and} \qquad \mathbf{v} = \langle v_1, v_2, v_3\rangle,$$

we also get

Parametric Equations for a Line. The standard parametrization of the line L through $P_0(x_0, y_0, z_0)$ parallel to $\mathbf{v} = \langle v_1, v_2, v_3\rangle$ is

$$x(t) = x_0 + tv_1, \qquad y(t) = y_0 + tv_2, \qquad \text{and} \qquad z(t) = z_0 + tv_3, \qquad \text{where} \qquad -\infty < t < \infty$$

Example. Find parametric equations for the line segment $\overline{PQ}$ connecting the points $P(-3, 2, -3)$ and $Q(1, -1, 4)$.

We are now equipped to compute the distance from a point to a line. Think of the problem in 2-space: suppose you have a line L that passes through the point P and is parallel to the vector $\mathbf{v}$. Take a point S that is not on the line. Notice that the distance from S to the line L is the length of the *perpendicular* we drop to form the vector-projection of $\overrightarrow{PS}$ onto $\mathbf{v}$:

Since $|\overrightarrow{PS} \times \mathbf{v}|$ computes the area of the paralellogram with sides $\overrightarrow{PS}$ and $\mathbf{v}$, we can use the fact that this area is equal to (base)(height) or ($|\mathbf{v}|$)(distance from S to L). Therefore,

$$\text{(Distance from a Point } S \text{ to a Line Through } P \text{ Parallel to } \mathbf{v}) = d(S, L) = \frac{|\overrightarrow{PS} \times \mathbf{v}|}{|\mathbf{v}|}$$

Example. Find the distance from the point $S(1, 1, 5)$ to the line L given by

$$x(t) = 1 + t, \qquad y(t) = 3 - t, \qquad \text{and} \qquad z(t) = 2t, \qquad \text{where} - \infty < t < \infty$$

An Equation for a Plane in Space

We know that the defining characteristic for a plane M is a normal vector $\mathbf{n} = \langle A, B, C \rangle$. Thus, if we know that the plane M contains the point $P_0(x_0, y_0, z_0)$, then any other point $P(x, y, z)$ in the plane M must exist so that the vector $\overrightarrow{P_0P}$ is orthogonal to $\mathbf{n}$.

That is, $\mathbf{n} \cdot \overrightarrow{P_0P} = 0$. Unpacking this,

$$\langle A, B, C \rangle \cdot \langle x - x_0, y - y_0, z - z_0 \rangle = 0.$$

Thus, the plane M consists of those points (x, y, z) where

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0.$$

Oftentimes, we will simplify the above equality to get

$$Ax + By + Cz = D, \qquad \text{where} \qquad D = Ax_0 + By_0 + Cz_0$$

Example. Find an equation for the plane going through $A(0, 0, 1)$, $B(2, 0, 0)$, and $C(0, 3, 0)$

Lines of Intersection

Just as lines are parallel if and only if they have the same slope (or 'tilt'), two planes are parallel if and only if their normal vectors are parallel – that is, if they are scalar multiples of one another. Two planes that are not parallel will always intersect in a line.

Example. Consider the planes M and N given by $3x - 6y - 2z = 15$ and $2x + y - 2z = 5$, respectively.

(i) Find a vector parallel to the line of intersection of the planes.

(ii) Find parametric equations for the line in which the planes intersect.

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We can also ask at what point a given line passes through a plane:

Example. Find the point where the line L, given by

$$x(t) = \frac{8}{3} + 2t, \qquad y(t) = -2t, \qquad \text{and} \qquad z(t) = 1 + t, \qquad \text{where} - \infty < t < \infty$$

intersects the plane $3x + 2y + 6z = 6$.

The Distance from a Point to a Plane

Let M be a plane with normal vector $\mathbf{n}$ containing the point P. Let S be any point *not* on M. Then the distance from S to the plane M is equal to the *scalar* projection of $\overrightarrow{PS}$ onto $\mathbf{n}$:

Thus, $$(\text{Distance from } S \text{ to the plane } M) = d(S, M) = \left| \overrightarrow{PS} \cdot \frac{\mathbf{n}}{|\mathbf{n}|} \right|$$

Example. Find the distance from $S(1, 1, 3)$ to the plane $3x + 2y + 6z = 6$.

Angles Between Planes

Lastly, we note that the angle between two intersecting planes is defined to be the acute angle between their normal vectors.

Example. Find the angle between the planes $3x - 6y - 2z = 15$ and $2x + y - 2z = 5$.